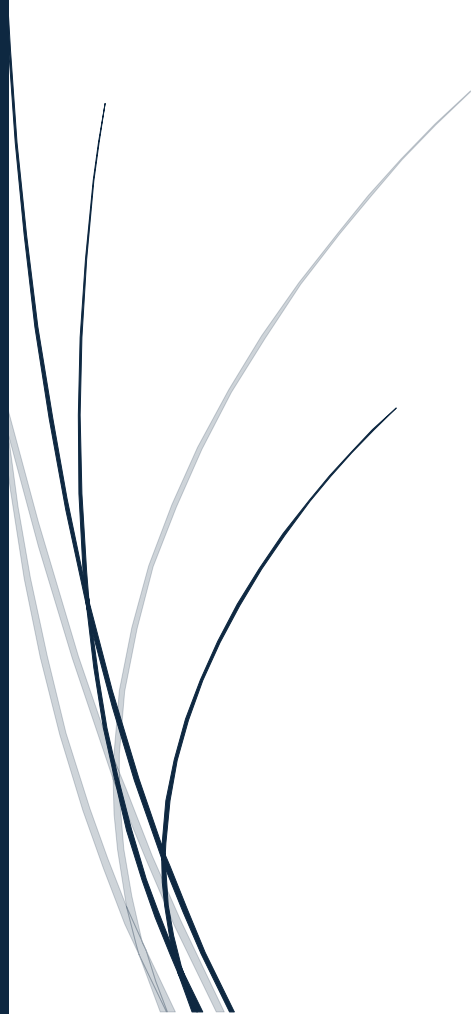
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Project plan

Personal project



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SEMESTER 7

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1. Introduction

Modern enterprises are increasingly adopting hybrid, multi-cloud or multi-site strategies to provide high availability and scalability. This can create multiple architectures problems, which requires a thoughtful planning of provisioning and security.

My idea of the project is adopt of an idea of a banking system application that require hardened infrastructure with a fast disaster recovery option. For that I wanted to use a two Kubernetes clusters located on two different sites (Netlab and Homelab) and use two different enterprise level Kubernetes deployment approach. After that connect the cluster, harden them with the advanced cybersecurity principles and setup a disaster recovery.

2. Research questions and project objectives.

Main Research Question: **How can we architect a secure enterprise infrastructure with Kubernetes for a banking system that provides an automated disaster recovery system?**

Sub Research Questions:

What is the current optimal way to architect and create enterprise level clusters?

What security considerations we need to think about while setting up and maintaining the Kubernetes clusters?

How can network and application-level policies prevent unauthorized lateral movement between workloads?

How can we automate the disaster recovery system so they are almost none downtime when the main cluster goes down?

Project objective is to design, implement, and validate a hybrid Kubernetes platform where workload communication is entirely identity-driven, secrets are centralized, and infrastructure is declarative. The ultimate objective is to prove the architecture's resilience by simulating lateral movement in the cluster and simulating a primary cluster failure.

3. Project Scope and Deliverables

In Scope:

- Deployment of two distinct Kubernetes clusters in different locations.
- Configuration of secure site-to-site connectivity between clusters.
- Implementation of Zero-Trust security policies.
- Implementation of automated disaster recovery.
- Deployment of a dummy banking application for testing purposes.

Out of Scope:

- Development of the actual banking application code.
- Synchronous multi-site database clustering.
- Hardware-level network configuration outside of the Kubernetes and VPN overlay layers.

Expected Deliverables:

- Final research report detailing the findings, DOT methodology, and evaluations.
- Architecture design and diagrams visualizing the Kubernetes environment.
- A GitHub repository containing all Infrastructure as Code and GitOps deployment manifests.
- Technical report explain all technical attributes of the project.
- A final presentation and live demonstration of the security and failover mechanisms.

4. Methodology

For the project I will use the DOT Framework to complete the research.

1. Available product analysis (Library)

I will use this method to map out the optimal ways of deployment of the enterprise level of the Kubernetes clusters. I will see what industry is currently for the underlaying software and hardware and choose an optimal stack for the project.

2. IT architecture sketching (Workshop)

Then I will architect the two clusters. I will map out the locations of the clusters itself as the applications, connectivity strategies, security hardening and the disaster recover.

After that I will create the project according to the design created in this method.

3. System and Security test (Lab)

After the realisation part is finished, I will try first System tests and cluster accessibility, application flow and finally disaster recovery to the second cluster. For the security part I will try to hack into the nodes and also test lateral movement within the cluster.

4. Peer review (Showroom)

Lastly I will showcase the project to my semester coaches and peers in the group project to get final feedback and analysis. After that I will draft a final report and close the project.

5. Non-functional requirements (scalability, security, reliability, performance).

Security: Unauthorized lateral movement must be prevented. Workload communication must be authenticated and encrypted via mTLS. Centralized secrets management must be utilized. Zero-trust principles must be used.

Reliability & Disaster Recovery: The architecture must survive a total loss of the primary cluster. Clear Recovery Time Objectives and Recovery Point Objectives must be defined, executed via state backups and declarative syncs to the secondary cluster.

Performance: The integration of security mechanisms and cross-site mesh routing must introduce only an acceptable, measured latency overhead.

Scalability: The centralized management plane must allow for the seamless addition of new nodes, workloads, or entirely new clusters without breaking the zero-trust policies.

Maintainability: All infrastructure and security configurations must be reproducible, version-controlled, and deployed via GitOps to eliminate configuration drift.

Observability: Security events, blocked traffic, and infrastructure metrics must be centrally visible and measurable.

6. Planning, timeline, and expected challenges.

Proposed Timeline:

- **Weeks 1-3:** Research and First Design.
- **Weeks 4-7:** Finish up the Design and start implementing.
- **Weeks 8-9:** Two cluster fully functional, ready to move to security hardening.
- **Weeks 10-12:** Security hardening and test applications deployment.
- **Weeks 13-15:** Implement Disaster Recovery, Backups.
- **Weeks 16-17:** Security and Resilience Testing.
- **Weeks 18-20:** Evaluation & Documentation.

Expected Challenge:

- **Cross-Site Networking Restrictions:** Connecting the Netlab to a Homelab over the public internet often introduces NAT, strict firewall, and routing challenges.
    - **Mitigation:** I will utilize a VPN solution designed for complex NAT traversal and allocate additional time for network debugging.
- **Heterogeneous Cluster Management:** Utilizing two different enterprise deployment approaches creates configuration overhead.
    - **Mitigation:** I will rely on GitOps and a centralized management planer to abstract OS-level differences and ensure declarative deployments across both sites.
- **Stateful Data Recovery:** Failing over stateless web frontends is simple, but failing over a stateful banking database without data corruption is highly complex.
    - **Mitigation:**. I will mitigate this by using asynchronous (as synchronous might be too unstable), scheduled state backups to a shared storage target, accepting a slightly higher but documented Recovery Point Objective.